

SOFTWARE ENGINEERING LAB 1

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AIML, G SECTION

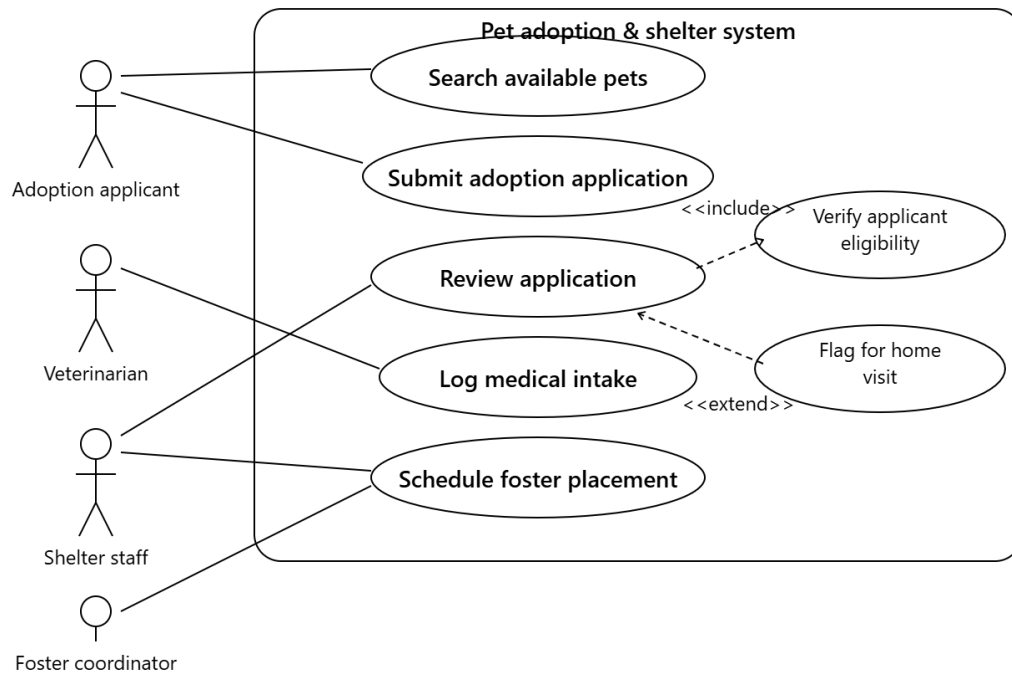
DELIVERABLE 1: REQUIREMENTS TABLE

Requirements Table — Pet Adoption & Shelter Management System

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow staff to log animal vaccination and medical intake records, updating animal availability status to 'Ready for Adoption'.	High	Pass: Adoption application linked to verified animal record. Fail: Animal under medical quarantine marked adopted.	Ensures adopters only receive medically cleared animals.
FR-002	Functional	The system shall allow an adoption applicant to submit an adoption application for a specific animal, including personal details and home environment responses.	High	Pass: Application is saved with a unique ID, linked to the animal record, and status is set to 'Pending Review'. Fail: An application missing a required field is not saved.	Gives shelter staff a structured, complete record to evaluate each applicant.
FR-003	Functional	The system shall allow shelter staff to review a submitted adoption application and approve or reject it.	High	Pass: Approving an application sets its status to 'Approved' and notifies the applicant. Fail: An application for an animal already marked 'Adopted' cannot be approved.	Prevents a single animal from being adopted by more than one applicant.
FR-004	Functional	The system shall allow shelter staff to schedule and track a foster home placement timeline for a rescued animal.	Medium	Pass: Entering a foster start date automatically computes and displays a review/end date. Fail: A negative or past start date is rejected with an error message.	Supports the scenario's foster placement tracking requirement and follow-up scheduling.

FR-005	Functional	The system shall allow an adoption applicant to search and filter available animals by species, age, breed, and compatibility with children.	Medium	Pass: Filtered results show only animals matching all selected criteria. Fail: An animal under medical quarantine appears in 'Ready for Adoption' search results.	Connects the browsing experience directly to accurate, up-to-date availability status.
NFR-001	Nonfunctional	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids) with page response under 200 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Keeps the browsing experience responsive even with a large, growing animal database.
NFR-002	Nonfunctional	The system shall restrict access to medical intake records and applicant personal data to authenticated shelter staff accounts only.	High	Pass: Unauthenticated or applicant-role users attempting to access medical/applicant records receive an access-denied response. Fail: Any endpoint returns medical or applicant PII without a valid staff session token.	Medical histories and applicant personal information are sensitive and must be protected from unauthorized access.

\\DELIVERABLE 2: UML USE-CASE DIAGRAM



DELIVERABLE 3: USE-CASE FLOW DOCUMENT

Use Case: Review Adoption Application

Primary Actor: Shelter Staff

Secondary Actor: Adoption Applicant

Preconditions

- ≠ The applicant has submitted a complete adoption application for a specific animal.
- ≠ The application status is 'Pending Review' and the linked animal is not already adopted.

Postconditions

- ≠ The application status is updated to 'Approved' or 'Rejected'.
- ≠ The linked animal's availability status is updated to reflect the decision.

Main Success Scenario

1. Shelter staff opens the list of pending adoption applications.
2. Staff selects an application to review.
3. System displays the applicant's details, home environment responses, and the linked animal record.
4. System automatically verifies applicant eligibility (include: Verify Applicant Eligibility) against basic screening rules.
5. Staff reviews the eligibility result together with the home environment responses.
6. Staff approves the application.
7. System updates the application status to 'Approved' and the animal status to 'Adoption Pending'.
8. System sends a confirmation notification to the applicant.
9. Use case ends successfully.

Extension Point

At step 5, if the home environment responses raise a concern (for example, no fenced yard for a large dog), staff may trigger the extending use case "Flag for Home Visit" (extend) before making a decision.

Alternate Flow — Application Rejected

- 6a1. Staff selects 'Reject' and enters a reason.
- 6a2. System updates the application status to 'Rejected' and reverts the animal's status to 'Ready for Adoption'.
- 6a3. System notifies the applicant of the rejection along with the reason.
- 6a4. Use case ends.